Homework #2 for Geospatial Analysis & Remote Sensing (EDS223 - Fall 2026)

2026-10-23

```
# Load in Packages
library(tidyverse)
library(here)
library(sf)
library(tmap)
library(gt)

# Load in data
ej <- sf::st_read(here::here(
  "data",
  "ejscreen",
  "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"
))

crashes <- sf::st_read(here::here(
  "data",
  "SWITRS_LA",
  "Crashes.csv"
))

map_inequal <- sf::st_read(here::here(
  "data",
  "mapping-inequality",
  "mapping-inequality-los-angeles.json"
))

# Filter EjScreen data
California_ej <- ej |>
  dplyr::filter(ST_ABBREV == "CA")
```

```r
# Filter dataset to a county
LA_county_ej <- California_ej |>
  dplyr::filter(CNTY_NAME == "Los Angeles County")
```

```r
# Check coordinate systems for compatibility
st_crs(LA_county_ej)
st_crs(crashes)
st_crs(map_inequal)

# Check to see CRS's match
if (st_crs(map_inequal) == st_crs(LA_county_ej)) {
  print("it's a match!")
} else {
  warning("still not a match")
}

# Quick test to see
st_crs(LA_county_ej) == st_crs(map_inequal)

# Change coorindate systems
map_inequal_transform <- st_transform(map_inequal, crs = st_crs(LA_county_ej))

# Check to see CRS's match
if (st_crs(map_inequal_transform) == st_crs(LA_county_ej)) {
  print("it's a match!")
} else {
  warning("still not a match")
}
```

```r
# Color dictionary for colors that accurately represent the grade level
colors_HOLC <- c(
  "A" = "forestgreen",
  "B" = "dodgerblue3",
  "C" = "goldenrod3",
  "D" = "firebrick3"
)
```

Part 1. Legacy of redlining in current environmental (in)justice

1. Create a map of historical redlining neighborhoods

```
# Map of Legacy Redlining in LA County

# Specifying dataframe from filtered data and
tm_shape(map_inequal_transform) +

  # Specify shape and customize specific attributes to enhance aesthetic
  tm_polygons(
    fill = "grade",
    lwd = 0.2,
    fill.legend = tm_legend(title = "Redline Grade"),
    fill.scale = tm_scale(values = colors_HOLC)
  ) +

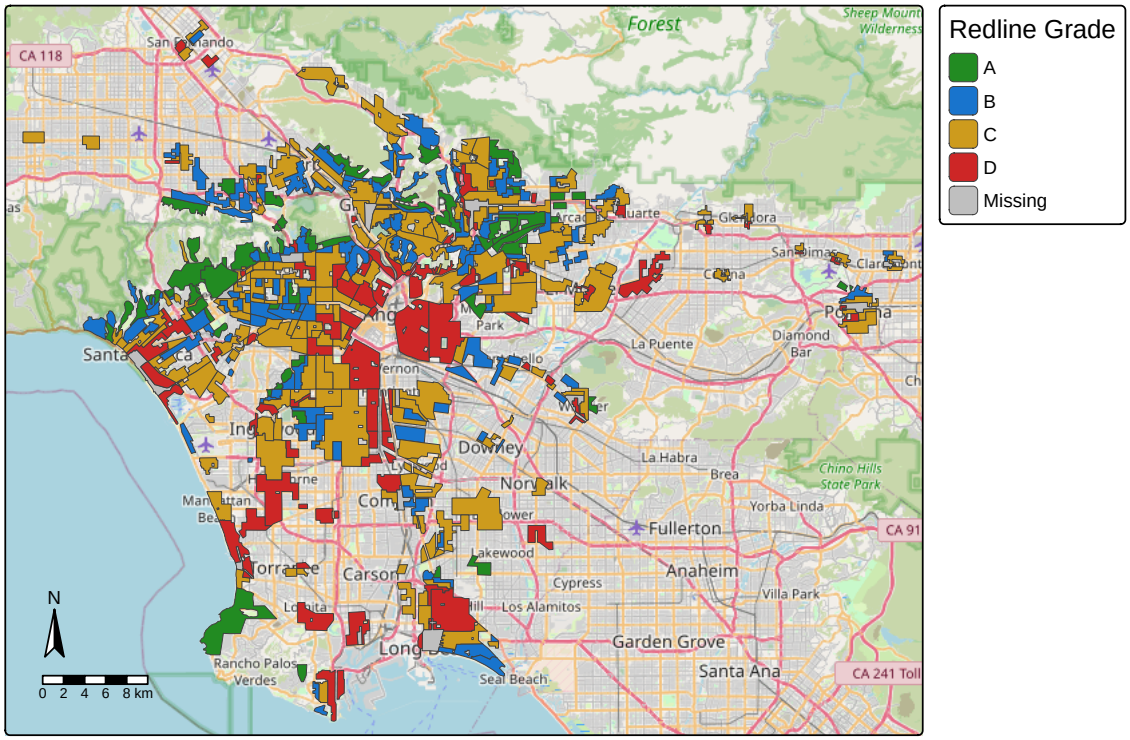
  # Adding an appropriate basemap
  tm_basemap("OpenStreetMap") +

  # Add a north arrow in the bottom left corner
  tm_compass(size = 2, text.size = 0.7, position = c("left", "bottom")) +

  # Adding a scalebar in the bottom left corner
  tm_scalebar(width = 8, text.size = 0.5, position = c("left", "bottom")) +

  # Add a title above the map, positioned to the left
  tm_title(
    "HOLC Redlining Grades in Los Angeles County",
    size = 1.0,
    position = tm_pos_out("center", "top", pos.h = "left", pos.v = "bottom")
  )
```

HOLC Redlining Grades in Los Angeles County



2. Create a table

```
# Perform left join on the datasets
df_joined <- st_join(LA_county_ej, map_inequal_transform) |>

  # Remove geometry to reduce computational load
  st_drop_geometry()


# Count the block groups
block_percent <- df_joined |>
```

Percentage of census block groups within each HOLC grade

In LA County

Grade	Percent
A	5.00
B	13.79
C	34.02
D	14.98
NA	32.22

```
  # Group by grade
  count(grade) |>

  # Calculate percent and round to two decimal places
  mutate(Percent = round(n / sum(n) * 100, 2)) |>

  # Rename "grade" column to "Grade"
  rename(Grade = grade) |>

  # Remove N column
  select(-n)
```

```
# Create gt summary table
block_percent |>

  # Pipe in gt summary table function
  gt() |>

  # Customize title and subtitle
  tab_header(
    title = "Percentage of census block groups within each HOLC grade",
    subtitle = md("*In LA County*")
  )
```

3. Create at least two visualizations summarizing current conditions

```r
# Find and graph the mean of LOWINCPCT

# Using the previously joined DataFrame
df_joined |>

  # Grouping by HOLC grade
  group_by(grade) |>

  # Calculating the variable of interest's mean and converting to %
  summarize(mean_pct = mean(LOWINCPCT, na.rm = TRUE) * 100) |>

  # Dropping NA values'
  drop_na(grade) |>

  # Piping in ggplot
  ggplot(aes(x = grade, y = mean_pct, fill = grade)) +

  # Specifying plot geometry
  geom_col(col = "black") +

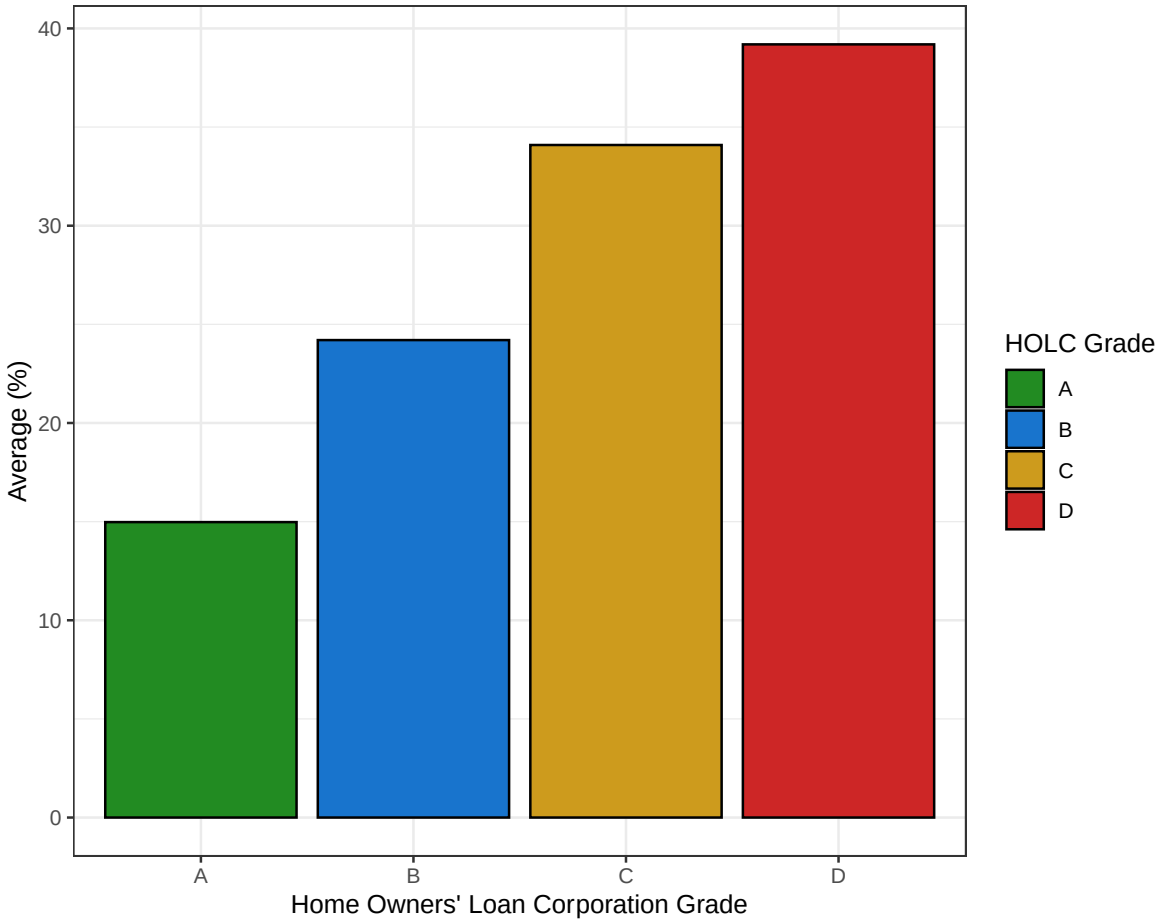
  # Customizing color scale
  scale_fill_manual(values = colors_HOLC) +

  # Setting theme
  theme_bw() +

  # Customizing titles
  labs(
    title = "Average Percentage of Low Income Living in HOLC Areas",
    x = "Home Owners' Loan Corporation Grade",
    y = "Average (%)",
    fill = "HOLC Grade",
    subtitle = "Figure 1"
  )
```

Average Percentage of Low Income Living in HOLC Areas

Figure 1




```
# Find and graph the mean of P_PM25

# Using the previously joined DataFrame
df_joined |>

  # Grouping by HOLC grade
  group_by(grade) |>

  # Calculating the variable of interest's mean
  summarize(mean_pct = mean(P_PM25, na.rm = TRUE)) |>

  # Dropping NA values'
  drop_na(grade) |>
```

```
# Piping in ggplot
ggplot(aes(x = grade, y = mean_pct, fill = grade)) +

# Specifying plot geometry
geom_col(col = "black") +

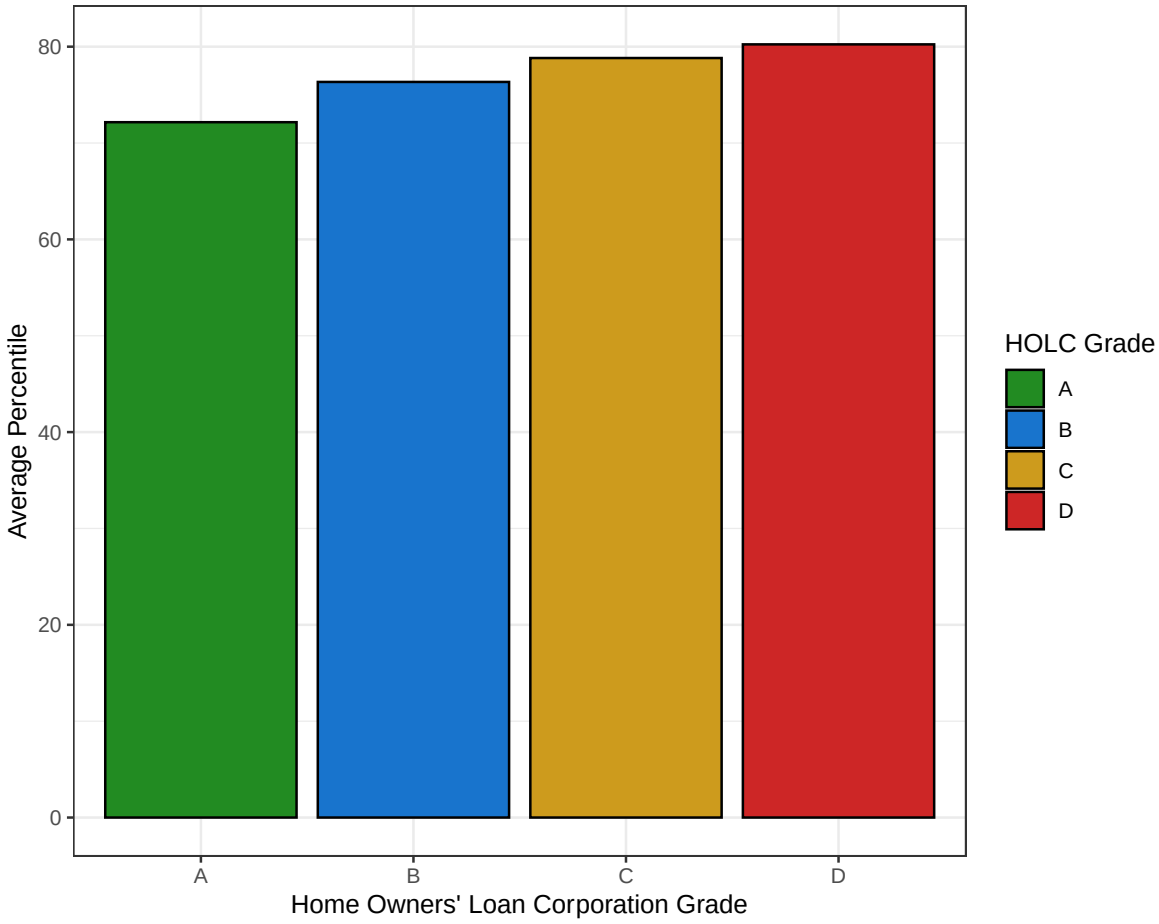
# Customizing color scale
scale_fill_manual(values = colors_HOLC) +

# Setting theme
theme_bw() +

# Customizing titles
labs(
  title = "Average Percentile for Particulate Matter 2.5 in HOLC Areas",
  x = "Home Owners' Loan Corporation Grade",
  y = "Average Percentile",
  fill = "HOLC Grade",
  subtitle = "Figure 2"
)
```

Average Percentile for Particulate Matter 2.5 in HOLC Areas

Figure 2




```
# Find and graph the mean of P_LIFEEXPPCT

# Using the previously joined DataFrame
df_joined |>

  # Grouping by HOLC grade
  group_by(grade) |>

  # Calculating the variable of interest's mean
  summarize(mean_pct = mean(P_LIFEEXPPCT, na.rm = TRUE)) |>

  # Dropping NA values'
  drop_na(grade) |>
```

```
# Piping in ggplot
ggplot(aes(x = grade, y = mean_pct, fill = grade)) +

# Specifying plot geometry
geom_col(col = "black") +

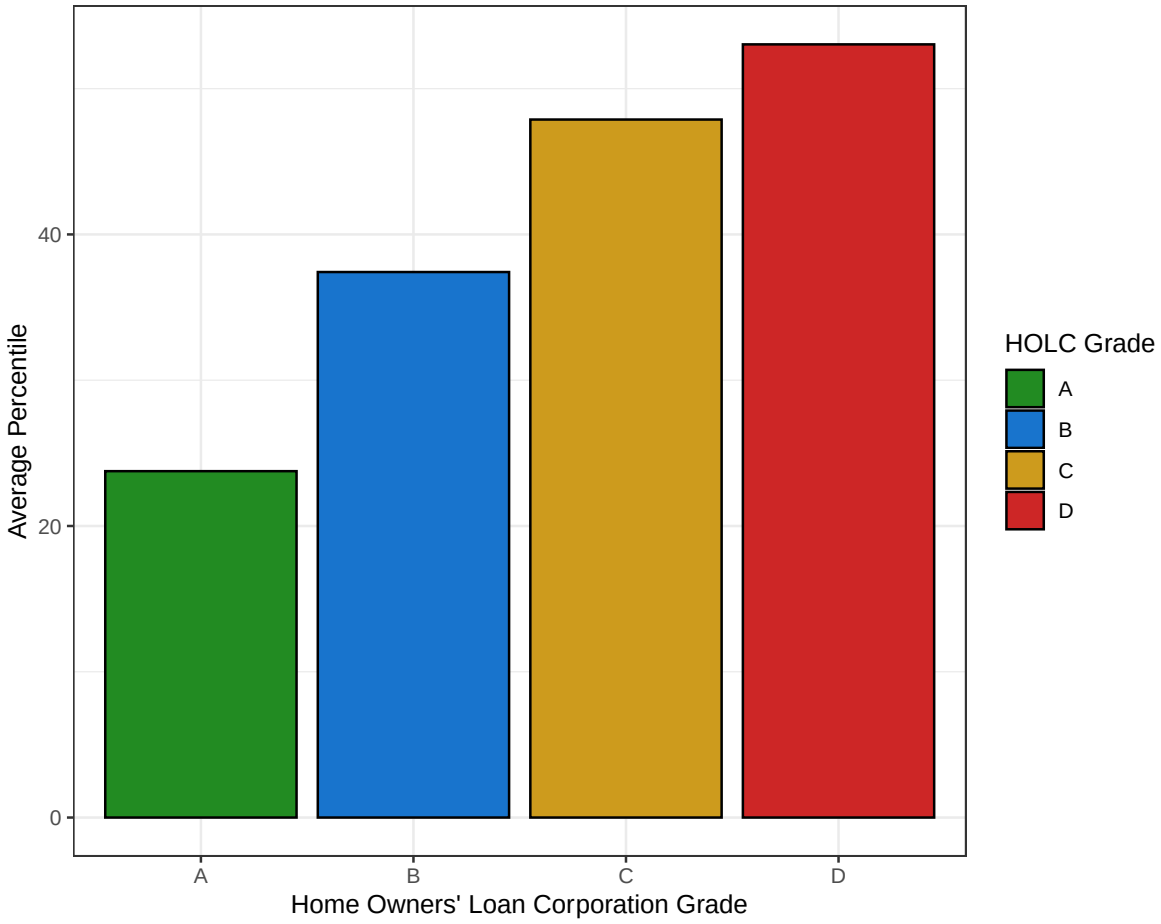
# Customizing color scale
scale_fill_manual(values = colors_HOLC) +

# Setting theme
theme_bw() +

# Customizing titles
labs(
  title = "Average Percentile for Low Life Expectancy in HOLC Areas",
  x = "Home Owners' Loan Corporation Grade",
  y = "Average Percentile",
  fill = "HOLC Grade",
  subtitle = "Figure 3"
)
```

Average Percentile for Low Life Expectancy in HOLC Areas

Figure 3



4. Write a brief paragraph reflecting on these results

Averaging the percent low income (LOWINCPCT), the PM2.5 percentile (P_PM25), and the low life expectancy percentile (P_LIFEEXPPCT) across census block groups in each HOLC grade shows a consistent pattern that conditions worsen steadily from Grade A to Grade D. In Figure 1, block groups in Grade D areas average nearly 40% low-income residents, compared with about 15% in Grade A areas. Figure 3 shows a similar gradient, with Grade D areas averaging around the 55th percentile for low life expectancy versus roughly the 25th percentile for Grade A. Figure 2 shows the same ordering for PM2.5, but the differences are much smaller. Every grade falls above the 70th percentile, which likely reflects poor air quality across Los Angeles county as a whole. Together, the results from these figures suggest that the effects of redlining persist nearly a century later. Neighborhoods rated worse than others were historically denied investment within the community, which made them targets for more highways and

industrial land uses. It is important to consider that these averages show association rather than causation, and they do not explicitly show the variation within each grade.

Part 2. Legacy of redlining in pedstrian and bike accidents

```
# Exploratory
unique(crashes$PEDESTRIAN_ACCIDENT)
unique(crashes$BICYCLE_ACCIDENT)
unique(crashes$ACCIDENT_YEAR)
unique(crashes$COUNT_BICYCLIST_KILLED)
unique(crashes$COUNT_PED_KILLED)


summary(crashes$POINT_X)
summary(crashes$POINT_Y)

summary(crashes$LATITUDE)
summary(crashes$LONGITUDE)

sum(crashes$LATITUDE == "NA")
sum(crashes$LONGITUDE == "NA")
sum(crashes$POINT_Y == "NA")
sum(crashes$POINT_X == "NA")
```

```
# Filter data
crash_filtered <- crashes |>

  # Select specific rows to reduce computational load
  select(
    COLLISION_DATE,
    BICYCLE_ACCIDENT,
    COLLISION_DATE,
    POINT_X,
    POINT_Y,
    ACCIDENT_YEAR,
    COUNT_PED_KILLED,
    COUNT_BICYCLIST_KILLED,
    PEDESTRIAN_ACCIDENT
  ) |>

  # Read date column as a date
```

```
  mutate(DATE = as.Date(COLLISION_DATE)) |>

  # Create year column
  mutate(YEAR = as.integer(year(DATE))) |>

  # Filter years
  filter(YEAR >= 2020 & YEAR <= 2025) |>

  # Filter to rows with accidents
  filter(PEDESTRIAN_ACCIDENT == "Y" | BICYCLE_ACCIDENT == "Y") |>

  # Read POINT_X column as a numerical value
  mutate(POINT_X = as.numeric(POINT_X)) |>

  # Read POINT_Y column as a numerical value
  mutate(POINT_Y = as.numeric(POINT_Y)) |>

  # Remove NA values'
  drop_na(POINT_X, POINT_Y) |>

  # Create coordinates from POINT_X and POINT_Y
  st_as_sf(coords = c("POINT_X", "POINT_Y"), crs = "EPSG:4326") |>

  # Change new DataFrame into having the same crs
  st_transform(st_crs(map_inequal_transform))
```

```
# Check to see if coordinate system matches
st_crs(crash_filtered) == st_crs(map_inequal_transform)
```

```
[1] TRUE
```

```
# Check to see CRS's match
if (st_crs(map_inequal_transform) == st_crs(crash_filtered)) {
  print("it's a match!")
} else {
  warning("still not a match")
}
```

```
[1] "it's a match!"
```

```r
# Join datasets
joined_crash_inequal <- st_join(crash_filtered, map_inequal_transform)
```

```r
# Create new DataFrame
plot_crash_inequal <- joined_crash_inequal |>

  # Remove geometry to reduce computational load
  st_drop_geometry() |>

  # Remove NA values'
  drop_na() |>

  # Group by HOLC grade
  group_by(grade) |>

  # Summarize total accidents and fatalities
  summarize(
    TOTAL = sum(
      PEDESTRIAN_ACCIDENT == "Y",
      BICYCLE_ACCIDENT == "Y",
      as.integer(COUNT_PED_KILLED) != 0,
      as.integer(COUNT_BICYCLIST_KILLED) != 0
    ),
    na.rm = TRUE
  ) |>

  # Turn total into a percent for each grade
  mutate(TOTAL_PCT = TOTAL / sum(TOTAL) * 100)
```

```r
# Plot based on total % of observations within redlined neighborhoods
ggplot(plot_crash_inequal, aes(x = grade, y = TOTAL_PCT, fill = grade)) +

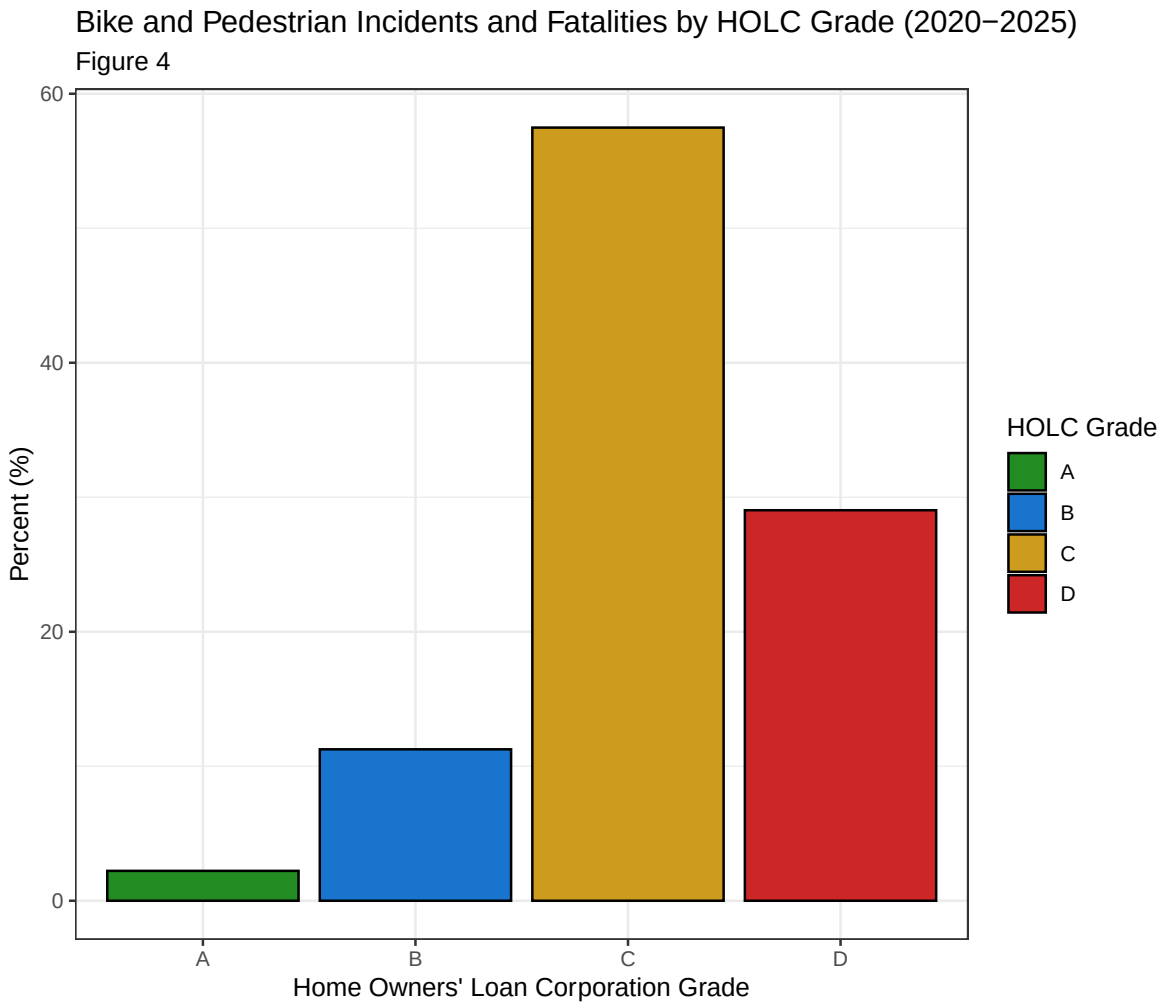
  # Specifying plot geometry
  geom_col(col = "black") +

  # Customzing color scale
  scale_fill_manual(values = colors_HOLC) +

  # Customizing theme
  theme_bw() +

  # Customizing titles
```

```
  labs(
    title = "Bike and Pedestrian Incidents and Fatalities by HOLC Grade (2020-2025)",
    subtitle = "Figure 4",
    x = "Home Owners' Loan Corporation Grade",
    y = "Percent (%)",
    fill = "HOLC Grade"
  )
```



Analysis: Figure 4 shows a clear disparity in bike and pedestrian crashes, including fatal crashes, across HOLC grades. Of all bike and pedestrian crashes recorded within HOLC-graded areas of Los Angeles County from 2020 to 2025, roughly 86% occurred in Grade C and D neighborhoods, while just over 2% occurred in Grade A neighborhoods. This pattern aligns with the long-term effects of redlining, where neighborhoods that were poorly rated were denied

funds for community investment, which resulted in inadequate infrastructure such as sidewalks, crosswalks, and bike lanes. As a result, residents of historically redlined neighborhoods face greater danger while partaking in everyday activities such as walking or riding a bike. Based on this analysis, further research into infrastructure and traffic conditions in these neighborhoods would help support targeted safety investments.